Python OOP Glossary

- **Object-Oriented Programming:** programming paradigm that organizes code around objects to model real-world entities and their interactions.
- **Object:** instance of a class that represents the specific data and behavior of an entity.
- **Instance:** a specific object created from a class.
- **Instantiation:** the process of creating an object from a class.
- **Instance attribute:** a variable associated with a specific object. This variable stores the data of the object and represents its characteristics.
- **Class attribute:** a variable that stores data about a class. Class attributes are shared by all instances of the class.
- **Method:** a function associated with an object, defining the actions that the object can perform or how it interacts with its data.
- **Constructor:** a special method that is automatically called to initialize the attributes of an object when it is created.
- **Abstraction:** the practice of only showing the essential features to the user, hiding the complexity of their internal implementation.
- **Encapsulation:** the practice of bundling data and methods within a class, like in a capsule that hides their internal implementation under a single unit.
- **Inheritance:** defining classes that inherit attributes from other classes.
- **Superclass (Parent class):** the class from which other classes inherit attributes and methods.
- **Subclass (Child class):** the class that inherits attributes and methods from another class.

- ✦ **Method overriding:** providing a specific implementation of a method in a subclass.

- ✦ **Method overloading:** defining multiple methods with the same name but different parameters. This is not directly supported by Python but it is common in object-oriented programming.

- ✦ **Polymorphism:** when objects of different classes can handle the same method class in different ways.

- ✦ **"Pythonic":** term commonly used to refer to an approach or syntax that follows the standards and best practices of Python.

- ✦ **Docstrings:** strings located at the top of a function, class, method, or module to describe its purpose, arguments, attributes, and important characteristics.

- ✦ **Mutation:** direct modification of an object's data within its allocated memory space.

- ✦ **Aliasing:** when multiple variable names (identifiers) reference the same memory location of an object.

- ✦ **Cloning:** the process of generating a new, distinct copy of an object in memory, with the same data as the original.